25901178

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCCMA201 - DIGITAL TOOLS FOR BUSINESS AND RESEARCH

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

SCHEME FOR VALUATION

Part A

Very Short Answer Questions Answer all questions

Each question carries **2** marks

1. **Explain why the interpretation of results is important in data analysis.** **[U] / [CO1]**

   (Write any two points).

   Helps understand results, Helps make correct decisions.

2. **How does data support decision making?** **[U] / [CO1]**

   Provides evidence, Reduces uncertainty.

3. **Define describing data numerically. Give two examples of numerical summary measures.**

   [K] / [CO2]

   Summarizing data using numbers.

   Examples: **Mean, Median**

4. **Name the different types of ANOVA.** **[K] / [CO3]**
   One-way ANOVA, Two-way ANOVA
5. **State the main difference between ANOVA and Kruskal-Wallis test.** **[U] / [CO3]**
   ANOVA is a parametric test that assumes normal distribution of data.
   Kruskal-Wallis test is a non-parametric test that does not assume normal distribution.

[2x5 = 10]

Part B

Short Answer Questions Answer any **5** questions

Each question carries **5** marks

6. **Explain how information supports decision making in business organizations.** **[U] / [CO1]**

(Definition of Information – 2 Marks; Role in Decision Making / Importance – 3 Marks)
Accurate data, Planning, Problem identification , Risk reduction , Performance improvement

7. **Explain how the SPSS/PSPP interface supports data analysis workflow.** **[U] / [CO1]**
   (***Overview of SPSS/PSPP Interface – 2 Marks; Components supporting workflow (Data View, Variable View, Output, Menu/Syntax) – 3 Marks)***
   Data View – Data entry
   Variable View – Define variables
   Menu Bar – Analysis
   Output Window – Results
   Syntax – Commands

8. **Explain the purpose of a line chart and a bar chart. How do they differ in the type of data they represent?** **[K] / [CO2]**

   (Purpose of Line Chart – 2 Marks; Purpose of Bar Chart – 2 Marks; Difference in Data Representation – 1 Mark)

Aspect	Line Chart	Bar Chart
Purpose	Shows trend	Shows comparison
Data Type	Continuous / Time data	Categorical data
Use	Changes over time	Compare categories
Representation	Points connected by lines	Rectangular bars
Main Difference	Trend	Comparison

9. **Compare Recoding and Visual Binning methods in SPSS. Explain when each method should be used.** **[U] / [CO2]**

   (Definition of Recoding – 2 Marks; Definition of Visual Binning – 2 Marks; When to Use Each Method – 1 Mark)

   Recoding – Change values/categories.
   Used to modify data values.
   Visual Binning – Create groups from ranges.

   Used for continuous data grouping.

10. **Illustrate with an example how regression is applied for prediction.** **[A] / [CO3]**

(Definition of Regression – 2 Marks; Example of Prediction using Regression – 3 Marks)
Regression predicts dependent variable.
Uses independent variable.

Example: Predict sales from advertising.

11. **Discuss the procedure for undertaking correlations.** **[U] / [CO3]**
(Definition of Correlation – 2 Marks; Steps / Procedure for Correlation Analysis – 3 Marks)
Enter data.
Select Analyze → Correlate → Bivariate.
Choose variables.
Select correlation method.
Click OK to get correlation value

12. **Interpret a Two-Way ANOVA output to identify main effects.** **[A] / [CO3]**
(Meaning of Two-Way ANOVA – 2 Marks; Identification of Main Effects from Output (using p-value / significance) – 3 Marks)
Check ANOVA table.
Identify factors.
Examine p-value.
$p < 0.05$ indicates significance.
Determine main effect of factors.

[5x5 = 25]

Part C

Essay Questions Answer any **1** questions

Each question carries **15** marks

13. **Explain skewness and kurtosis. Explain how these measures describe the shape of a distribution. Discuss Positive and negative skewness. How these measures support interpretation of data.** **[U] / CO2]**

(Definition of Skewness – 3 Marks; Definition of Kurtosis – 3 Marks; Shape of Distribution – 3 Marks; Positive & Negative Skewness – 3 Marks; Interpretation / Importance in Data Analysis – 3 Marks)

Skewness – Measure of symmetry
Kurtosis – Measure of peakedness
Positive Skewness – Tail to the right
Negative Skewness – Tail to the left
Use – Describes shape of distribution

14. **Describe null hypothesis, alternate hypothesis, significance level and p-value in respect of paired-sample t-test.** [U] / [CO3]

(Null Hypothesis – 4 Marks; Alternate Hypothesis – 3 Marks; Significance Level – 4 Marks; p-value – 4 Marks)

Null Hypothesis (H_0) – No difference
Alternate Hypothesis (H_1) – Difference

Significance Level (α) – 0.05 level
p-value – Probability value

Decision –
$p \leq 0.05$ Reject H_0 **[15x1 = 15]**
$p > 0.05$ Accept H_0